MLOW-Transformer for High-Energy Proton Flux Forecasting in Space Weather Time Series

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Key Points:

- A multivariate space weather forecasting framework is developed for short-term high-energy proton flux ($p10$) prediction using LSTM, Transformer, and MLOW-Transformer models.
- A low-rank frequency-domain decomposition (MLOW) is introduced to enhance the modeling of non-stationary proton flux variations and extreme events.
- The proposed MLOW-Transformer integrates frequency-domain decomposition with Transformer-based temporal modeling to improve prediction accuracy and peak capture capability.

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Abstract

Forecasting high-energy proton flux under non-stationary space weather conditions remains a significant challenge. To address this, we propose the MLOW-Transformer, which combines a low-rank frequency-domain decomposition module with a Transformer-based forecasting architecture. By applying Fourier transforms and low-rank frequency basis learning, the model successfully decomposes multivariate space weather time series into mean, dominant frequency, and residual components before utilizing the Transformer to capture deep temporal dependencies. Using a 12-hour historical window of solar wind speed, the Kp index, electron flux, and proton fluxes, the model predicts the subsequent 1-hour high-energy proton flux. Comparative evaluations reveal that the MLOW-Transformer yields superior performance over LSTM and standard Transformer models, particularly in overall prediction accuracy, the characterization of temporal dynamics, and peak response. Building on these predictions, we introduce an operational space weather warning scheme that categorizes continuous flux forecasts into four discrete alert levels (no alert, yellow, orange, and red). Our findings establish that coupling low-rank frequency decomposition with Transformer-based temporal modeling is a robust, data-driven approach for enhancing proton flux forecasting and advancing automated space weather warning systems.

Plain Language Summary

High-energy proton flux is an important indicator of space radiation conditions. When solar activity increases, proton flux can rise rapidly and affect satellites, spacecraft electronics, and space mission safety. Therefore, predicting its variation is essential for space weather monitoring and risk warning.However, proton flux exhibits strong sudden variations, rare extreme events, and complex temporal patterns, making accurate forecasting challenging for conventional models.To address this issue, we propose an MLOW-Transformer model. The model first extracts dominant frequency components from time series using MLOW-based low-rank frequency decomposition, and then applies a Transformer to learn the relationship between historical space weather observations and future proton flux.Experimental results show that the proposed model outperforms the baseline Transformer in prediction accuracy and better captures peak variations. In addition, the predicted results can be further converted into multi-level warning information, providing a practical tool for space weather radiation environment monitoring.

1 Introduction

Space weather refers to solar activity and the resulting variations in the Sun-Earth space environment, including solar flares, coronal mass ejections (CMEs), solar energetic particle (SEP) events, geomagnetic disturbances, and changes in the energetic particle environment (Temmer, 2021). With the development of satellite communication, navigation and positioning, spacecraft operation, and space exploration missions, human society has become increasingly dependent on the near-Earth space environment. Severe space weather events may cause damage to satellite electronic components, communication interruptions, increased navigation errors, and enhanced aviation radiation risks. Therefore, accurate forecasting of key space weather parameters is of great significance for space weather monitoring, risk assessment, and the construction of warning systems.

High-energy proton flux is an important indicator of the space radiation environment. In particular, proton flux with energy greater than 10 MeV is commonly used to evaluate solar proton events and their radiation risk levels. After solar eruptive activity occurs, high-energy proton flux may increase rapidly within a short period of time and pose potential threats to spacecraft, satellite payloads, and the safety of space ac-

tivities. However, high-energy proton flux remains at low levels most of the time and only shows sharp increases during a small number of disturbed periods. Such sparsity, abruptness, and imbalance make its short-term forecasting a challenging task (Posner, 2007; Papaioannou et al., 2022; Stumpo et al., 2021).

Solar energetic particle events are usually closely related to solar flares, CMEs, interplanetary shock propagation, and particle transport processes (Waterfall et al., 2023). Early studies mainly relied on physical indicators, empirical rules, or statistical relationships for prediction. For example, Posner (Posner, 2007) proposed a short-term warning method for solar energetic ion events using relativistic electrons. Papaioannou et al. (Papaioannou et al., 2022) proposed the probabilistic solar particle event forecasting model PROSPER. Stumpo et al. (Stumpo et al., 2021) analyzed the sample distribution and false-alarm problems in solar proton event prediction from a machine learning perspective. Waterfall et al. (Waterfall et al., 2023) investigated the relationships between high-energy solar particle events and the parameters of solar flares, CMEs, and ground-level enhancement events. These studies provide an important basis for the prediction of space energetic particle events. However, due to the complex nonlinear coupling among solar activity, solar wind propagation, geomagnetic response, and particle flux variations, it remains difficult to achieve stable and accurate flux forecasting by relying only on empirical rules or limited physical indicators.

In recent years, with the continuous accumulation of space weather observational data, machine learning and deep learning methods have gradually become an important direction in space weather forecasting. The FLARECAST project demonstrated the application potential of big data and machine learning techniques in solar flare prediction (Georgoulis et al., 2021). Park et al. (Park et al., 2018) applied deep convolutional neural networks to flare prediction using full-disk solar magnetograms. Sun et al. (Sun et al., 2023) constructed deep learning-based flare forecasting models using solar observation images. Lavasa et al. (Lavasa et al., 2021) compared the performance of multiple machine learning methods in SEP event prediction. Sadykov et al. (Sadykov et al., 2021) compared machine learning-based solar proton event prediction results with operational forecast results. Torres et al. (Torres et al., 2025) used historical electron and proton observations to predict future high-energy SEP proton flux. These studies indicate that data-driven models can learn complex nonlinear relationships from historical observations and are suitable for processing multivariate and high-dimensional space weather data. Nevertheless, the rarity of extreme events, non-stationarity of time series, imbalanced data distributions, and limited model interpretability still restrict the application of deep learning models in high-reliability warning scenarios.

From the modeling perspective, high-energy proton flux forecasting belongs to the category of multivariate time series forecasting. Future variations in $p10$ are not only related to its own historical state, but may also be affected by variables such as solar wind speed, Kp index, electron flux, and low-energy proton flux. Solar wind speed reflects the propagation state of the solar wind, the Kp index characterizes the level of global geomagnetic disturbance, and electron flux and low-energy proton flux reflect changes in the space energetic particle environment at different energy ranges. Therefore, using multiple space weather parameters as input helps the model learn potential relationships among variables. At the same time, high-energy proton flux exhibits non-stationarity and long-tailed distribution characteristics. As a result, models tend to learn common low-value samples while underestimating abrupt peaks and high-value intervals.

Deep neural networks have shown strong capability in time series forecasting tasks. Long short-term memory (LSTM) networks alleviate the gradient vanishing problem in traditional recurrent neural networks through gated structures and can learn long-term dependencies within a certain range (Hochreiter & Schmidhuber, 1997). However, their sequential computation structure has limitations in long-sequence modeling and parallel efficiency. The Transformer model directly establishes dependencies among different

time steps through the self-attention mechanism and has strong parallel computing capability and long-range dependency modeling ability (Vaswani et al., 2017). Therefore, it has gradually been applied to time series forecasting tasks. For long-sequence and non-stationary time series, models such as Informer, Autoformer, FEDformer, Non-stationary Transformer, PatchTST, and TimesNet have made improvements from the perspectives of sparse attention, sequence decomposition, frequency-domain representation, non-stationary modeling, sequence patching, and multi-period modeling (H. Zhou et al., 2021; Wu et al., 2021; T. Zhou et al., 2022; Liu et al., 2022; Nie et al., 2023; Wu et al., 2023). These studies have promoted the development of time series forecasting models from single time-domain modeling toward decomposition modeling, frequency-domain modeling, and multi-scale modeling.

Although Transformer has strong temporal modeling ability, the standard Transformer mainly learns historical sequence dependencies from the time domain. For non-stationary sequences such as space weather data, important variation patterns may appear not only in the time domain but also in the frequency-domain structure. Low-frequency components usually reflect long-term background variations, while high-frequency components may correspond to short-term disturbances or abrupt fluctuations. Studies on Fourier transform and FEDformer have shown that frequency-domain representation can enhance the model's ability to describe long-term trends and periodic structures (Cooley & Tukey, 1965; T. Zhou et al., 2022). The MLOW method further learns the main frequency patterns in a sequence through low-rank decomposition and decomposes the original sequence into a mean component, low-rank frequency-domain components, and residual information (Yang et al., 2026). This makes it suitable for characterizing the complex features of high-energy proton flux, in which background variations and local disturbances coexist.

Based on the above background, this study proposes an MLOW-Transformer model for short-term high-energy proton flux forecasting. First, LSTM and Transformer models are constructed as baseline methods for performance comparison. Then, a low-rank frequency-domain decomposition method (MLOW) is introduced, which decomposes the input sequence into mean components, low-rank frequency-domain components, and residual information before being fed into the Transformer. This design enhances the model's ability to capture non-stationary dynamics and frequency-domain structures.The main contributions of this study are as follows:

- A multivariate space weather-based forecasting framework for high-energy proton flux prediction;
- An MLOW-Transformer model that integrates frequency-domain decomposition with Transformer-based temporal modeling;
- A systematic comparison among LSTM, baseline Transformer, and MLOW-Transformer under the same forecasting setting;
- A space weather warning framework that converts continuous predictions into discrete alert levels, including no alert, yellow, orange, and red warnings.

The rest of this paper is organized as follows. We first describe the dataset and preprocessing procedure and define the forecasting problem. Then, we present the architectures of the LSTM, Transformer, and MLOW-Transformer models. Next, experimental results and comparisons are provided, followed by analysis of forecasting performance and warning effectiveness. Finally, conclusions and future work are discussed.

2 Method

This study designs a forecasting method for high-energy proton flux in multivariate space weather time series. The overall framework includes data preprocessing, sliding-window sample construction, a baseline Transformer forecasting model, an MLOW-Transformer

forecasting model, and a space weather warning system based on the prediction results. Different from general time series forecasting tasks, high-energy proton flux has obvious non-stationarity, long-tailed distribution, and abrupt peak characteristics. Therefore, this study focuses not only on the model's ability to fit the overall variation trend, but also on its ability to identify high-value intervals and warning levels.

Five key space weather variables are selected as model inputs, including solar wind speed, Kp geomagnetic activity index, electron flux, low-energy proton flux, and high-energy proton flux. The space weather state vector at time step t is defined as

$$\boldsymbol{x}_t = [bulkspeed_t, kp_t, e2_t, p1_t, p10_t] \in \mathbb{R}^C \tag{1}$$

where $bulkspeed_t$ denotes solar wind speed, kp_t denotes the geomagnetic activity index, $e2_t$ denotes electron flux, $p1_t$ denotes low-energy proton flux, $p10_t$ denotes high-energy proton flux, and $C = 5$ denotes the number of input variables. Given a historical window with length L, the model input sequence is represented as

$$\boldsymbol{X}_{t-L+1:t} = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t] \in \mathbb{R}^{L\times C}. \tag{2}$$

The forecasting objective is to predict the high-energy proton flux value after H future time steps based on the previous L time steps of multivariate space weather observations:

$$\hat{\boldsymbol{y}}_{t+H} = f_\theta(X_{t-L+1:t}), \tag{3}$$

where f_θ denotes the forecasting model, θ denotes model parameters, and $\hat{y}_{t+H}$ denotes the predicted future high-energy proton flux. This study mainly uses data with a 1 hr temporal resolution. The input window length is set to $L = 12$, and the prediction horizon is set to $H = 1$, which means that the previous 12 hr of space weather observations are used to predict the high-energy proton flux $p10$ in the next 1 hr.

In the data preprocessing stage, the 1 hr space weather data obtained by maximum-value resampling are first loaded. Solar wind speed, Kp index, electron flux, low-energy proton flux, and high-energy proton flux are then selected to construct the multivariate input sequence. Since electron flux, low-energy proton flux, and high-energy proton flux have large numerical ranges and obvious long-tailed distributions, a base-10 logarithmic transformation is applied to these three particle flux variables:

$$x' = \log_{10}(x). \tag{4}$$

The logarithmic transformation compresses the order-of-magnitude differences in particle flux, reduces the influence of extreme values on model training, and preserves the relative variations in high-value intervals. To eliminate differences in dimensions and numerical ranges among variables, each input variable is further standardized:

$$z = \frac{x - \mu}{\sigma}, \tag{5}$$

where x denotes the original variable value, μ and σ denote the mean and standard deviation of the corresponding variable, and z denotes the standardized value. After standardization, different space weather variables are converted to relatively consistent numerical scales, which is beneficial for stable neural network training.

After data preprocessing, a sliding-window method is used to construct supervised learning samples. Given the historical input length L and the prediction length H, the input sequence of the i-th sample is

$$\boldsymbol{X}_i = [x_i, x_{i+1}, \ldots, x_{i+L-1}] \tag{6}$$

The corresponding prediction target is

$$\boldsymbol{Y}_i = [y_{i+L}, y_{i+L+1}, \ldots, y_{i+L+H-1}] \tag{7}$$

By moving the window step by step along the time axis, the continuous space weather time series is transformed into multiple input-output sample pairs. To preserve the temporal consistency of the forecasting task, the data are divided into training, validation, and test sets in chronological order, with proportions of 70%, 15%, and 15%, respectively. The training set is used for model parameter learning, the validation set is used to monitor generalization ability and convergence during training, and the test set is used for final performance evaluation.

To compare the performance of different deep learning models in high-energy proton flux forecasting, an LSTM model was first constructed as a recurrent neural network baseline. LSTM controls the retention and updating of historical information through input, forget, and output gates, enabling it to capture temporal dependencies in sequential data. For the input feature vector $\mathbf{x}_t$ at time step t, the hidden state and cell state of the LSTM can be expressed as

$$(\boldsymbol{h}_t, \boldsymbol{c}_t) = \mathrm{LSTM}(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}, \boldsymbol{c}_{t-1}), \tag{8}$$

where $\boldsymbol{h}_t$ and $\boldsymbol{c}_t$ denote the hidden state and cell state at time step t, respectively. In this study, a multivariate space weather sequence covering the previous 12 hr was fed into the LSTM network, and the hidden state at the final time step was used to predict the future high-energy proton flux:

$$\hat{y}_{t+H} = \boldsymbol{W}_o \boldsymbol{h}_t + \boldsymbol{b}_o, \tag{9}$$

where $\boldsymbol{W}_o$ and $\boldsymbol{b}_o$ denote the weight matrix and bias vector of the output layer, respectively. The LSTM model consisted of two LSTM layers followed by a fully connected prediction layer, which produced the 1 hr-ahead high-energy proton flux forecast. To ensure a fair comparison, the LSTM, Transformer, and MLOW-Transformer models used the same input variables, sliding-window configuration, chronological data split, and evaluation metrics.

A baseline Transformer model is first constructed as the comparison benchmark. The baseline Transformer takes the multivariate historical sequence generated by the sliding window as input. The input sequence first passes through a linear mapping layer, which maps the original five-dimensional variables into a high-dimensional feature space:

$$\boldsymbol{Z}_t = \phi(\boldsymbol{x}_t \boldsymbol{W}_1 + \boldsymbol{b}_1), \tag{10}$$

where $\boldsymbol{W}_1$ and $\boldsymbol{b}_1$ are the parameters of the linear mapping layer, and $\phi(\cdot)$ denotes the nonlinear activation function. The features are then further mapped to a hidden space with dimension d_{model}. In this study, $d_{\mathrm{model}} = 256$.

Since the self-attention mechanism in Transformer does not contain temporal order information by itself, positional encoding is added to the input features. The positional encoding is defined as

$$\begin{cases} PE(pos, 2i) = \sin\left(\dfrac{pos}{10000^{2i/d_{\mathrm{model}}}}\right), \\ PE(pos, 2i+1) = \cos\left(\dfrac{pos}{10000^{2i/d_{\mathrm{model}}}}\right). \end{cases} \tag{11}$$

where *pos* denotes the time-step position and i denotes the hidden-dimension index. By adding positional encoding, the model can use both observed values and temporal order information, allowing it to distinguish the influence of different time points in the historical window on the prediction target.

The core of the Transformer Encoder is the self-attention mechanism. For the input feature matrix $\mathbf{X}$, the model first obtains the query, key, and value matrices through linear transformations:

$$\begin{cases} \boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}^Q, \\ \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}^K, \\ \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}^V. \end{cases} \tag{12}$$

The self-attention mechanism is calculated as

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V}, \tag{13}$$

where d_k denotes the dimension of the key vectors. This mechanism assigns attention weights according to the relationships among different historical time steps, enabling the model to automatically focus on important historical information when predicting future high-energy proton flux. After the Transformer Encoder, the temporal features are flattened and passed through fully connected prediction layers to output the high-energy proton flux forecast for the next 1 hr. Figure 1 shows the architecture of the baseline Transformer model.

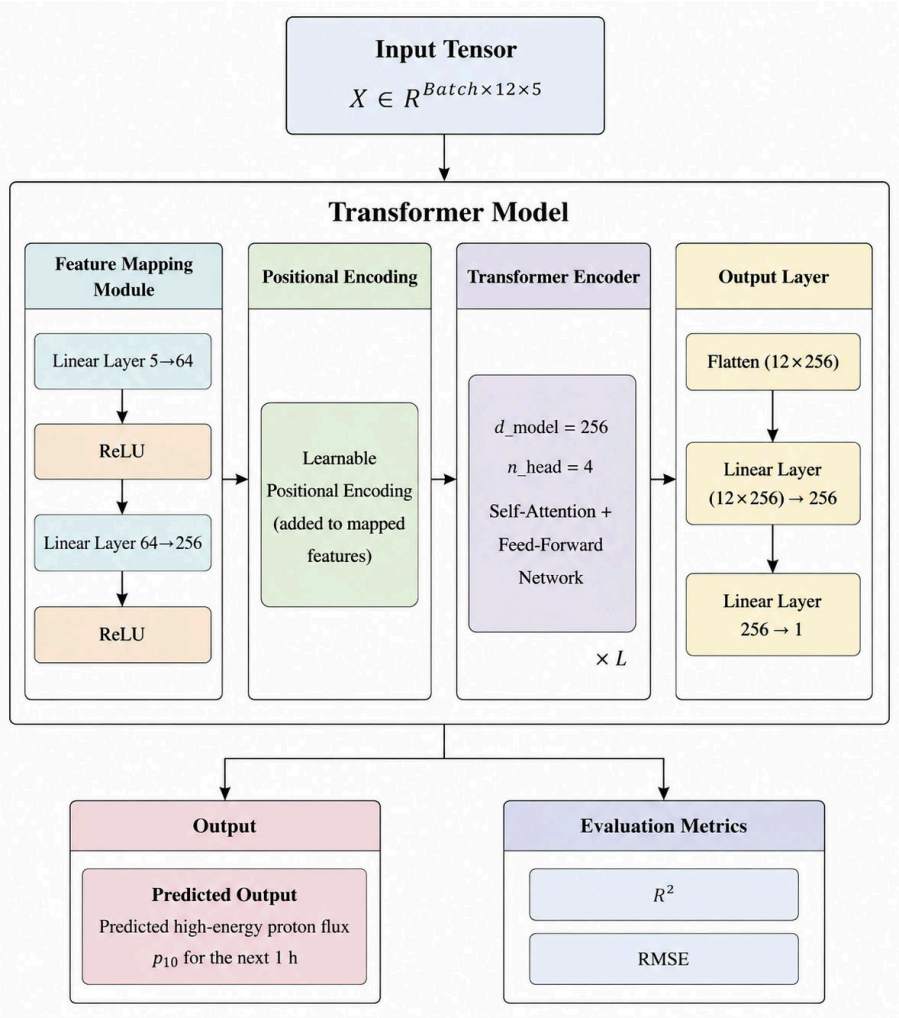


Figure 1: Architecture of the baseline Transformer model for high-energy proton flux forecasting.

The baseline Transformer can learn temporal dependencies in multivariate space weather time series, but its input is still the original time-domain sequence. For non-stationary signals such as high-energy proton flux, the sequence often contains slowly varying background components, dominant frequency-varying components, and short-term disturbance

residuals. If only time-domain modeling is used, the model may not fully distinguish variation patterns at different time scales and may easily underestimate abrupt peaks and extreme high values. Therefore, this study introduces the low-rank frequency magnitude decomposition of multiple effects (MLOW) method based on the baseline Transformer and constructs the MLOW-Transformer model.

The core idea of MLOW-Transformer is to perform low-rank frequency-domain decomposition on the input time series before Transformer modeling. For an input sequence $\boldsymbol{X} \in \mathbb{R}^{B\times S\times F}$, where B denotes the batch size, S denotes the input sequence length, and F denotes the number of variables, the fast Fourier transform is first applied along the temporal dimension:

$$\boldsymbol{F} = \mathrm{FFT}(\boldsymbol{X}). \tag{14}$$

For a univariate discrete sequence with length S, $x_0, x_1, \ldots, x_{S-1}$, its discrete Fourier transform can be written as

$$X_k = \sum_{n=0}^{S-1} x_n e^{-j2\pi kn/S}, \quad k = 0, 1, \ldots, S-1. \tag{15}$$

where X_k denotes the complex coefficient corresponding to the k-th frequency component, and j denotes the imaginary unit. The amplitude of the Fourier coefficient reflects the contribution of different frequency components in the original sequence.

After obtaining the frequency-domain representation, MLOW extracts the amplitude information of non-zero frequency components and performs low-rank decomposition on the frequency amplitude matrix. Let the frequency amplitude matrix be $\boldsymbol{M}$. MLOW approximates it as

$$\boldsymbol{M} \approx \boldsymbol{A}\boldsymbol{H}, \tag{16}$$

where $\boldsymbol{A}$ denotes the coefficient matrix of samples on the low-rank frequency basis, and $\boldsymbol{H}$ denotes the learned low-rank frequency basis. To obtain a more stable and representative frequency basis, this study first uses principal component analysis to initialize the frequency amplitude information and then further optimizes the low-rank frequency basis through hyperplane non-negative matrix factorization. The learned $\boldsymbol{H}$ matrix is saved after training and remains unchanged during validation, testing, and real-time prediction, ensuring that a consistent frequency decomposition basis is used at different stages.

After MLOW decomposition, the original input sequence is decomposed into the mean component, low-rank frequency-domain components, and residual information. The overall form is represented as

$$\boldsymbol{X} = \boldsymbol{X}_{\mathrm{mean}} + \sum_{r=1}^{R} \boldsymbol{X}_r + \boldsymbol{X}_{\mathrm{res}}. \tag{17}$$

where $\boldsymbol{X}_{\mathrm{mean}}$ denotes the mean component, which reflects the background level in the historical window; $\boldsymbol{X}_r$ denotes the r-th low-rank frequency-domain component; R denotes the decomposition rank; and $\boldsymbol{X}_{\mathrm{res}}$ denotes the residual information used to retain local disturbances that cannot be explained by the dominant frequency components. In this study, the rank is set to $R = 2$. Therefore, each input sample after MLOW processing contains the mean component, two low-rank frequency-domain components, and the residual component.

In the MLOW-Transformer model, the mean component is retained separately, while the low-rank frequency-domain components and residual component are sent into the subsequent neural network for prediction. Let the decomposed input be

$$\boldsymbol{X}_{\mathrm{MLOW}} \in \mathbb{R}^{B\times(R+2)\times S\times F}, \tag{18}$$

where $R+2$ corresponds to the mean component, R low-rank frequency-domain components, and the residual component. The model first extracts the mean component corresponding to the target variable at the last time step as the baseline value:

$$m = \boldsymbol{X}_{\text{mean}}(S, p_{10}). \tag{19}$$

Then, the low-rank frequency-domain components and residual information are expanded along the feature dimension to form sequence features suitable for Transformer input:

$$\boldsymbol{Z} \in \mathbb{R}^{B\times S\times((R+1)F)}. \tag{20}$$

Here, $R+1$ denotes three low-rank frequency-domain components and one residual component, and $F = 5$ denotes the number of variables. Then, $\boldsymbol{Z}$ is processed by linear mapping, positional encoding, and the Transformer Encoder to extract temporal dependencies from the decomposed sequence. Finally, the model obtains the predicted residual through fully connected layers and adds it to the retained mean component:

$$\hat{y}_{t+H} = g_\theta(\boldsymbol{Z}) + m, \tag{21}$$

where $g_\theta(\cdot)$ denotes the output of the Transformer prediction network. This design enables the model to use both the variation components after frequency-domain decomposition and the background level in the historical window, thereby better adapting to the non-stationary variation of high-energy proton flux. Figure 2 shows the architecture of the MLOW-Transformer model.

Compared with the baseline Transformer, the main difference of MLOW-Transformer lies in its input representation. The baseline Transformer directly uses the original time-domain sequence and mainly relies on self-attention to learn relationships among time steps. In contrast, MLOW-Transformer first performs low-rank frequency-domain decomposition before inputting the sequence into the Transformer. The complex sequence is decomposed into the background component, dominant frequency components, and residual component, and the Transformer then learns temporal dependencies from the decomposed sequence. Therefore, MLOW-Transformer combines frequency-domain decomposition with Transformer time-domain modeling, which helps improve the representation of non-stationary space weather sequences.

The mean squared error (MSE) is adopted as the loss function for model training. Given a batch of N samples, the loss is defined as:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{22}$$

Model parameters are optimized using the Adam optimizer. The Transformer baseline is trained for 80 epochs with early stopping based on validation loss, while the MLOW-Transformer is trained for 50 epochs. After training, the models are evaluated on the test set using the coefficient of determination and root mean squared error.

The coefficient of determination is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}. \tag{23}$$

where $\bar{y}$ denotes the mean of ground truth values. A higher R^2 indicates better explanatory power of the model.

The root mean squared error (RMSE) is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}. \tag{24}$$

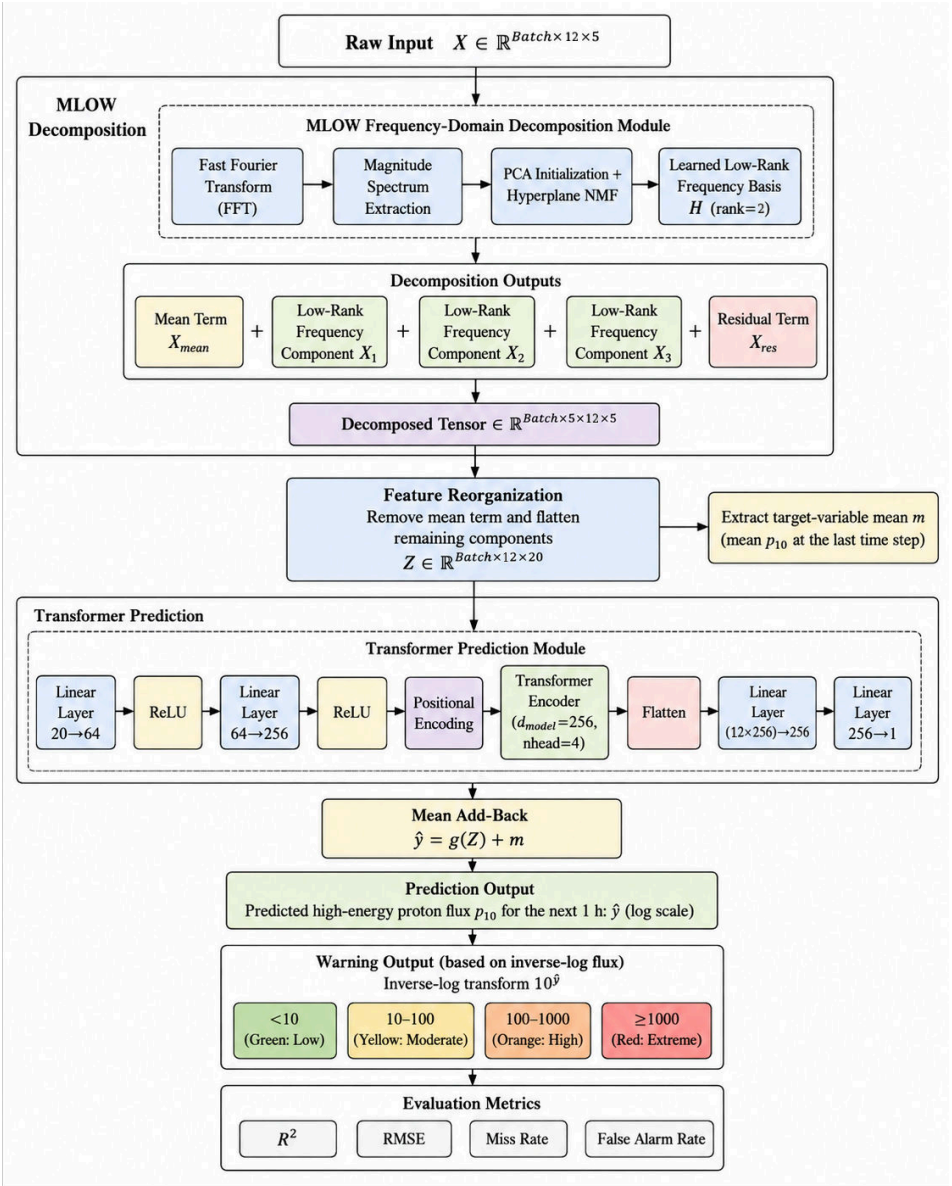


Figure 2: Architecture of the MLOW-Transformer model for high-energy proton flux forecasting.

Due to the large magnitude variations in proton flux, model training and evaluation are conducted in logarithmic space, and results are later transformed back to the original scale for visualization and warning analysis.

A space weather warning system is constructed based on the predicted high-energy proton flux. The model output in logarithmic scale is transformed back to the original scale:

$$\hat{p}_{10} = 10^{\hat{y}}. \tag{25}$$

The restored $\hat{p}_{10}$ is then used to determine the warning level at the future time step. According to the predicted high-energy proton flux, this study sets three warning thresholds. The warning level is defined as

$$A(\hat{p}_{10}) = \begin{cases} 0, & \hat{p}_{10} < 10, \\ 1, & 10 \le \hat{p}_{10} < 100, \\ 2, & 100 \le \hat{p}_{10} < 1000, \\ 3, & \hat{p}_{10} \ge 1000. \end{cases} \tag{26}$$

Here, 0 denotes no warning, 1 denotes yellow warning, 2 denotes orange warning, and 3 denotes red warning.

To evaluate the effectiveness of the warning system, this study further compares the true high-energy proton flux levels with the predicted warning levels. Let the true warning level be A_i, and the predicted warning level be $\hat{A}_i$. When the true level is not lower than yellow warning but the predicted level is no warning, it is regarded as a missed warning. When the predicted level is lower than the true level, it is regarded as underprediction. When the predicted level is higher than the true level, it is regarded as a false warning. The missed warning rate, underprediction rate, and false warning rate are defined as

$$\begin{cases} R_{\text{miss}} = \dfrac{\sum_{i=1}^{N} \mathrm{I}(\hat{A}_i = 0,\ A_i \ge 1)}{\sum_{i=1}^{N} \mathrm{I}(A_i \ge 1)}, \\ R_{\text{under}} = \dfrac{\sum_{i=1}^{N} \mathrm{I}(\hat{A}_i < A_i,\ A_i \ge 1)}{\sum_{i=1}^{N} \mathrm{I}(A_i \ge 1)}, \\ R_{\text{false}} = \dfrac{\sum_{i=1}^{N} \mathrm{I}(\hat{A}_i > A_i,\ \hat{A}_i \ge 1)}{\sum_{i=1}^{N} \mathrm{I}(\hat{A}_i \ge 1)}. \end{cases} \tag{27}$$

Here, $\mathrm{I}(\cdot)$ is the indicator function, which equals 1 when the condition is satisfied and 0 otherwise.

3 Discussion

This section compares the performance of LSTM, the baseline Transformer, and MLOW-Transformer for high-energy proton flux ($p10$) forecasting. The experiments used a 12 hr-to-1 hr sliding-window setting, where the previous 12 hr of multivariate space weather observations were used to predict the proton flux in the following 1 hr. Model performance was evaluated on both logarithmic and original scales to assess the ability of each model to reproduce the overall variations and extreme peaks.

Figures 3 show the LSTM prediction results on the logarithmic and original scales, respectively. The LSTM model captured the overall temporal variation of high-energy proton flux and showed good agreement with the observations during low-flux and relatively stable periods. However, the predicted curve became smoother during abrupt enhancement events, and several high peaks were underestimated.

Figures 4 present the prediction results of the baseline Transformer. The Transformer model also reproduced the general evolution of proton flux. Nevertheless, it showed

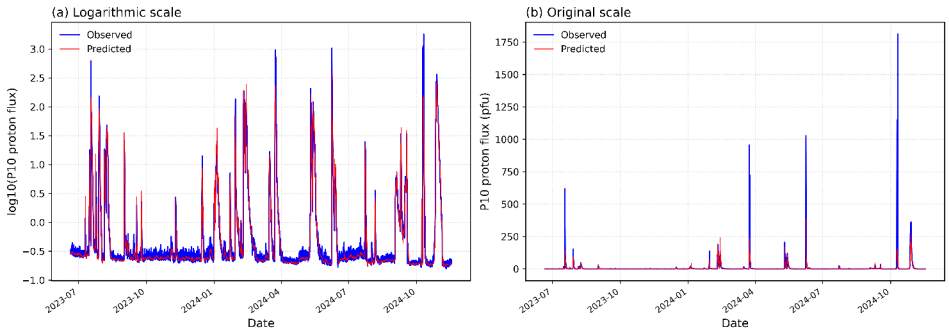


Figure 3: LSTM prediction results of high-energy proton flux

underestimation in several high-flux intervals, particularly during sharp and short-duration peaks. On the original scale, the predicted amplitudes of several extreme proton flux events were lower than the observed values.

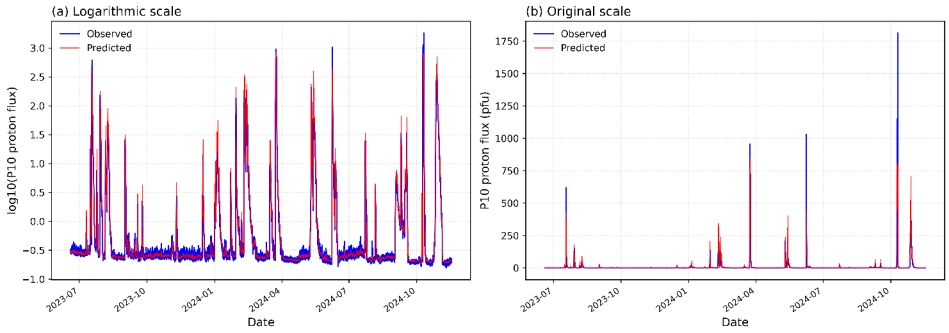


Figure 4: Transformer prediction results of high-energy proton flux.

Figures 5 show the prediction results of MLOW-Transformer. Compared with LSTM and the baseline Transformer, MLOW-Transformer showed better agreement with the observed proton flux in high-value intervals. It more accurately identified the occurrence times of major peaks and better followed the rising and declining phases of strong proton flux events. These results indicate that the MLOW decomposition improved the representation of non-stationary variations and enhanced the response of the model to abrupt changes.

The quantitative results further support the visual comparison. LSTM achieved an R^2 value of 0.9803 and an RMSE of 0.0924. The baseline Transformer achieved an R^2 value of 0.9785 and an RMSE of 0.0965. MLOW-Transformer obtained the best performance, with an R^2 value of 0.9841 and an RMSE of 0.0829. Therefore, MLOW-Transformer provided the highest overall explanatory capability and the lowest prediction error among the three models.

To further evaluate the capability of the MLOW-Transformer model in reproducing high-energy proton flux variations, we examined the prediction performance from two perspectives: overall prediction agreement and event-level peak estimation.

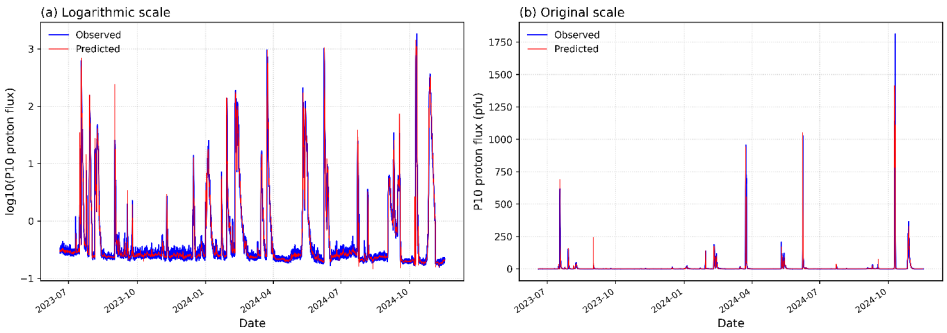


Figure 5: MLOW-Transformer prediction results of high-energy proton flux

Figure 6(a) presents the density distribution between the observed and predicted P10 proton fluxes. The predicted values show strong agreement with observations in logarithmic space, with most samples concentrated around the 1:1 reference line. The Pearson correlation coefficient reaches 0.992, demonstrating that the model effectively reproduces the overall variability of proton flux. However, larger deviations appear in the high-flux region, indicating remaining uncertainties in estimating extreme proton enhancement peaks.

Figure 6(b) compares the observed and predicted peak fluxes of major proton enhancement events. The results show that the MLOW-Transformer successfully reproduces most proton enhancement events, and the predicted peak fluxes generally agree well with observations. Nevertheless, several intense SEP events exhibit underestimated peak values, suggesting that accurately estimating extreme proton flux maxima remains challenging for the current model.

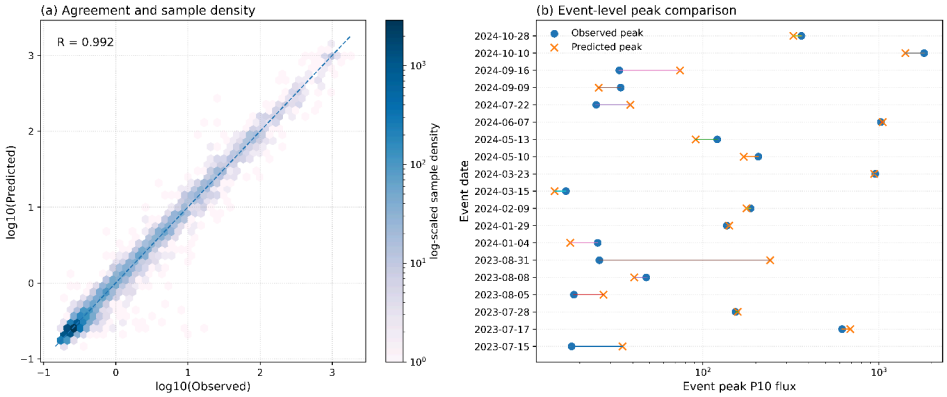


Figure 6: Prediction diagnostics of the MLOW-Transformer model. (a) Agreement between observed and predicted P10 proton fluxes. The color shading represents the logarithmic sample density, the dashed line indicates the 1:1 reference line, and R denotes the Pearson correlation coefficient. (b) Comparison of observed and predicted peak fluxes for major proton enhancement events. Blue circles and orange crosses represent observed and predicted peaks, respectively, with the x-axis shown on a logarithmic scale.

Based on the MLOW-Transformer predictions, a high-energy proton flux warning system was further developed. The predicted proton flux was divided into four warning

levels: no warning for $p10 < 10$, yellow warning for $10 \leq p10 < 100$, orange warning for $100 \leq p10 < 1000$, and red warning for $p10 \geq 1000$.

Figure 7 shows the observed and predicted proton flux together with the warning thresholds from June 2023 to November 2024. The warning system successfully captured most proton flux enhancement periods and provided timely responses for the major yellow, orange, and red warning events. The overall missed-warning rate was 4.83%, the underprediction rate was 8.56%, and the false-warning rate was 8.64%. The proportions of inaccurately predicted yellow, orange, and red warning events were 9.46%, 16.93%, and 50.00%, respectively. Overall, the warning system showed good capability for detecting yellow and orange warning events, while extreme red warning events remained challenging because of peak underestimation and insufficient warning-level prediction.

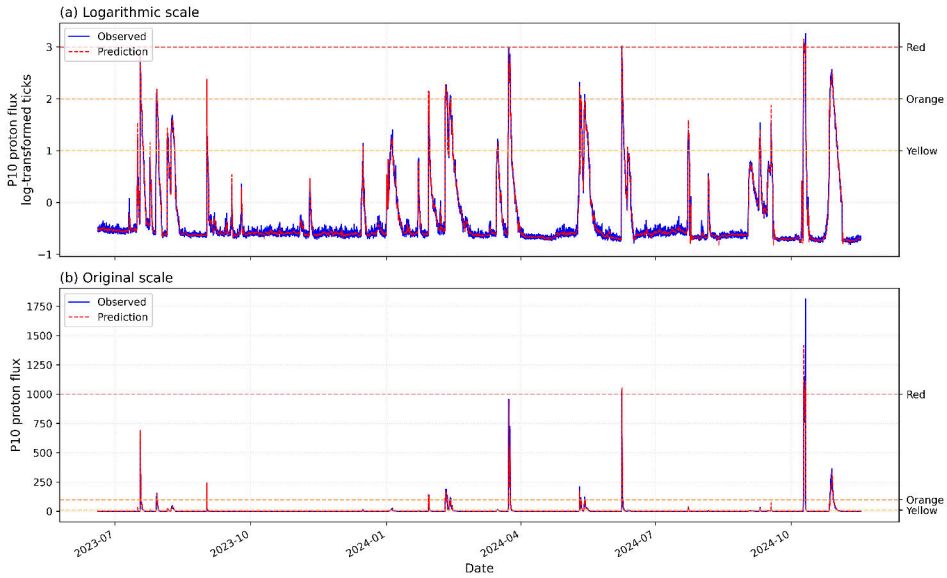


Figure 7: High-energy proton flux prediction and warning results based on MLOW-Transformer from June 2023 to November 2024. (a) prediction results in logarithmic scale; (b) prediction results in original scale. Dashed lines indicate the thresholds for yellow, orange, and red warning levels.

4 Conclusion

This study addresses the short-term forecasting problem of high-energy proton flux ($p10$) in multivariate space weather time series. Three models are developed, including LSTM, a baseline Transformer, and the proposed MLOW-Transformer. A warning system is further constructed based on the forecasting results. Experimental results show that both LSTM and the Transformer baseline can capture the overall temporal variations of $p10$. In contrast, the proposed MLOW-Transformer achieves better performance in modeling non-stationary dynamics and extreme fluctuations, leading to improved prediction accuracy, particularly for peak events. Based on the predicted results, the outputs of MLOW-Transformer are transformed back to the original scale and mapped into four warning levels: no warning, yellow, orange, and red alerts. This enables the conversion from continuous predictions to operational space weather warning information.

Overall, the MLOW-Transformer achieves the best performance among all compared models, demonstrating that the integration of low-rank frequency-domain decom-

position and Transformer-based temporal modeling is effective for space weather forecasting. However, limitations remain in the current study, including the scarcity of extreme event samples, underestimation of certain strong peaks, and limited incorporation of solar source information. Future work will consider integrating multi-source solar observations such as solar flares, coronal mass ejections, and active region magnetic field data, as well as improving extreme-event weighting strategies and warning level optimization.

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